

YASHVI VAGHELA

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EDUCATION

University of Southern California, Los Angeles

August 2023 - May 2025

Master of Science in Computer Science

(GPA 3.81/4)

Courses: Analysis of Algorithms, Database Systems, Applied Natural Language Processing, Web Technologies, Machine Learning for Data Science, Information Retrieval and Web Search Engines, Deep Learning and Its Applications.

University of Mumbai, Mumbai

August 2019 - May 2023

Bachelor of Engineering in Computer Engineering

(GPA 9.35/10)

Courses: Data Structures, Database Management System, Operating System, Object-Oriented Programming in Java, Python Programming, Artificial Intelligence, Cloud Computing, Machine Learning, Applied Data Science.

TECHNICAL SKILLS

- Programming Languages and Databases: Python, C++, Java, MySQL, MongoDB, SQL, Oracle, PostgreSQL, SQLite
- Web Technologies: HTML, CSS, JavaScript, TypeScript, ReactJs, NodeJs, Angular, Express, Django, Flask, RESTful APIs, FastAPI, Next.js
- Tools & Libraries: TensorFlow, OpenCV, Numpy, AWS, Git, Docker, Google App Engine, Figma, Data Visualization, LangChain, Qdrant, RAG, Vector Databases

EXPERIENCE

Easley-Dunn Productions Inc- Software Developer

July 2025 - May 2026

- Developed a video annotation tool with frame-accurate playback, scrubbing, and hotkey tagging, enabling efficient labeling and export of structured gameplay datasets in CSV format using React.js
- Integrated CV models into a Flask backend to process annotated gameplay data into move-probability graphs, heatmaps, and a real-time bounding-box overlay on the video feed.

Ikigai-The Art of Living, India- UX/UI Intern

June 2022 - July 2022

- Designed and prototyped responsive UI/UX components in Figma, including input controls, navigation elements, and informational layouts, optimizing user flows and enhancing overall site usability.
- Collaborated with developers to integrate designs into production code, reducing front-end bugs by 20%.

PROJECTS

StudyBot (Python, FastAPI, LangChain, Groq API, Qdrant, SQLite, Next.js, Gemini)

- Built a full-stack RAG-based study assistant using LangChain and Groq API to generate AI-based study plans and stream real-time chat responses, adapting explanations based on struggle patterns detected within each topic.
- Designed a FastAPI backend with SQLite and a vector DB for sub-500ms semantic retrieval, referencing struggling topics, detecting prerequisite gaps, and surfacing related concepts, with plans adjustable via natural conversation.

Stock Search Website (MongoDB, Express, Angular, NodeJs, GCP)

- Developed a stock search web application using Angular and Node.js, integrating Finnhub and Polygon APIs to display real-time market data for 50+ stocks.
- Implemented portfolio simulation features including buy/sell transactions and watchlists, with user data stored in MongoDB, backend services deployed on Google App Engine, and caching added for frequently requested ticker data to reduce redundant API calls and improve response consistency.

FitGear-Gym Ecommerce Website (MongoDB, Express, ReactJs, NodeJs)

- Built a full-stack e-commerce site for browsing and purchasing 50+ gym equipment products, using React and Node.js/Express, with fitness blogs, embedded videos, and simulated checkout via a dummy Paytm API.
- Designed MongoDB schemas and backend APIs for product, user, and order data, and integrated an interactive React chatbot for BMI calculation, contributing to a ~35% increase in user engagement.

Interview Feedback Assistant (Html, CSS, Flask, Python, NLP)

- Built an interview feedback platform that analyzed recorded responses and generated automated feedback using NLP pipelines.
- Implemented resume-job description similarity scoring with TF-IDF and cosine similarity, and used spaCy to detect speech patterns and undesirable phrases in interview responses.

RESEARCH PAPERS

Implementation of Applied Machine Learning for Identification of Exoplanets Using NASA's API

- Developed and evaluated ML models on NASA's Kepler dataset for exoplanet classification, achieving 91% accuracy with Random Forest after SMOTE-based class balancing; findings were published in the International Journal of Design Engineering (2021).

EXTRA CURRICULAR ACTIVITIES

- Top 10/50+ teams in Rubix Hackathon' 23. Led a team of 3 to build a blog website where users could read others' blogs, get recommendations, publish content, and apply paraphrasing, grammar, and spell-check features.